



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions on spending patterns, customer segments, and subscription behavior.

Dataset at a Glance

3,900

Purchases

Transactional records analyzed

18

Columns

Features per record

37

Missing Values

In Review Rating column

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Previous Purchases, Frequency, Review Rating,
Shipping Type

Data Preparation in Python

01

Load & Explore

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean Missing Data

Imputed missing Review Rating values using the **median rating per product category**.

03

Standardize & Engineer

Renamed columns to snake_case; created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`.

04

Load to PostgreSQL

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.



SQL Analysis — Revenue & Spending

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

Revenue by Gender

Compared total revenue generated by male vs. female customers.

High-Spending Discount Users

Identified customers who used discounts but still spent **above the average purchase amount**.

SQL Analysis — Products & Shipping

Top 5 Products by Rating

Found products with the highest average review ratings.

Shipping Type Comparison

Compared average purchase amounts between **Standard** and **Express** shipping.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Subscribers vs. Non-Subscribers

Compared **average spend** and **total revenue** across subscription status to understand the value of subscriber segments.

SQL Analysis — Segments & Categories

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Customer Segmentation

Classified customers into **New**, **Returning**, and **Loyal** segments based on purchase history.

Top 3 Products per Category

Listed the most purchased products within each category.

Discount-Dependent Products

Identified **5 products** with the highest percentage of discounted purchases.

SQL Analysis — Repeat Buyers & Age Groups

Repeat Buyers & Subscriptions

Checked whether customers with >5 **purchases** are more likely to subscribe.

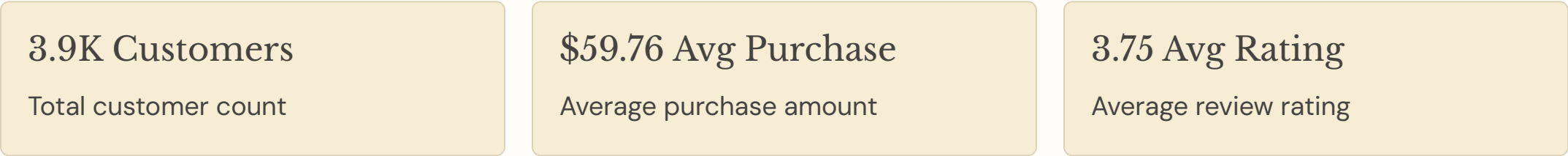
Revenue by Age Group

Calculated total revenue contribution of each age group.

	age_group  text	total_revenue  numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763



Power BI Dashboard



Interactive filters for **Subscription Status**, **Gender**, **Category**, and **Shipping Type**. Charts cover revenue and sales by category and age group, plus subscription distribution (27% Yes, 73% No).

Business Recommendations

→ Boost Subscriptions

Promote exclusive benefits for subscribers.

→ Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

→ Review Discount Policy

Balance sales boosts with margin control.

→ Product Positioning

Highlight top-rated and best-selling products in campaigns.

→ Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

Thank You!

We appreciate your time and interest in our customer shopping behavior analysis. Let's drive growth together.